

Diagnostics



Special Tool(s)

 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Diagnosing an electronically controlled automatic transmission is simplified by using the following procedures. The most important things to remember is there is a definite procedure to follow. Do not take shortcuts or assume that critical checks or adjustments have already been made. Follow the procedures as written to avoid missing critical components or steps. By following the diagnostic sequence, the technician can diagnose and repair the concern the first time.

On-Board Diagnostic (OBD) with Scan Tool

These self-tests should be used to diagnose the PCM and should be carried out in order.

NOTE: Make sure the latest software version is loaded on the scan tool.

1. If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

NOTE: The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC to the VCM .

2. If the scan tool does not communicate with the VCM :
 - check the VCM connection to the vehicle.
 - check the scan tool connection to the VCM .
 - refer to Section 418-00, No power to the scan tool.
3. If the scan tool does not communicate with the vehicle:
 - verify the ignition is in the RUN position.
 - verify scan tool operation with a known good vehicle.
 - refer to Section 418-00, The PCM does not respond to the scan tool.
4. Carry out the network test:
 - If the scan tool responds with no communication from one or more modules, refer to Section 418-00, No High Speed Controller Area Network (CAN) communication, all modules are not responding.
 - If the network test passes, retrieve and record the continuous memory DTCs .
5. Clear the continuous DTCs and carry out the PCM self-test.
6. If DTCs retrieved are related to the concern, GO to the DTC Chart.
7. If no DTCs related to the concern are retrieved, REFER to Diagnosis By Symptom in this section.